8.6.4 Numerical Example

Let us briefly summarize the numerical experiments of [220] using the SIN(R) method to simulate a system of 512 flexible water molecules using a fully flexible molecular model [294] and the smooth particle-mesh Ewald method (SPME) [123] (see Appendix A) to compute electrostatic forces. Initial simulations were conducted without a fast-slow force decomposition to demonstrate the effectiveness of the method as a thermostatting scheme. This technique was shown to allow accurate computation of the OH, HH and OO radial distribution functions when suitably small timesteps were used.

As explained in Chap. 4, the resonances in multiple timestepping restrict the size of the large (outer) stepsize usable in simulation. In the case of fully flexible water the maximum outer stepsize that can be used for stable RESPA simulation is typically found to be less than 5 fs [245, 329]. A multiple timestepping method for the SPME method can be based on the decomposition

$$U_{\text{Total}} = U_{\text{bonds}} + U_{\text{s.r.}} + U_{\text{l.r.}}$$

where U_{bonds} includes length and angle bonds, $U_{\text{s.r.}}$ includes all other short-ranged components, and $U_{\text{l.r.}}$ incorporates both the "reciprocal space part" from Particle Mesh Ewald summation (normally calculated using 3D FFT) as well as longer-ranged screened Coulombic pair terms within the simulation cell. Then a nested multiple timestepping hierarchy can be used with stepsizes h_{inner}, h_{mid} and h_{outer}, each larger stepsize being an integer multiple of the next smaller. There is no difficulty with adapting the SIN(R) scheme (which was presented for a two-level setup) to this multi-level hierarchical timestepping framework.

The article [220] provides more detail of the SPME parameters and a detailed analysis of the numerical performance. We merely mention here that with $h_{\text{inner}} = 0.5\,\text{fs}$ and $h_{\text{mid}} = 3\,\text{fs}$, it was possible to use outer stepsizes of up to 99 fs for stable integration and accurate resolution of all three radial distribution functions, with no evidence of resonances. The short-range cutoff in the reported simulations was around 8Å. High efficiency improvements would be anticipated in simulations with a large simulation cell. These methods could be valuable in more complex applications, including systems with mixed quantum and classical simulation domains, and in the setting of high performance computing architectures, e.g. GPU systems.

8.7 Generalized Langevin Dynamics

We have already encountered Generalized Langevin dynamics in Chap. 6, where it appeared as an intermediate stage in the derivation of Langevin dynamics. In some cases, the properties of the heat bath and the resulting memory kernel are important